B2 — DPO and IPO on a language model: task, goal, and how it was solved

What phase B2 asks, why a preference loss can stand in for B1's RL loop, where that shortcut breaks, and what the runs showed. The lab notebook with the numbers is `experiments/b2_dpo/README.md`; every table and figure here is produced by `experiments/b2_dpo/report.py` from the run directories.

> **At a glance**
>
> - **DPO matches PPO's gold score at PPO's KL budget for a tenth of the compute.** On B1's task, with pairs labelled by the reward model PPO trained against, DPO at β = 0.2 reaches gold P(positive) 0.957 ± 0.007 at 9.6 nats of KL (3 seeds, ≈5 min per run on a T4); B1's best PPO reaches 0.973-0.983 at 11-14 nats (≈55 min per run). Gold rises with KL at the same rate for both; PPO buys more *reward-model* score per nat (4.7 against 3.0), which the gold judge does not credit.
> - **DPO learns the reward model's exploit from the labels, partially.** Chosen responses contain Chinese 2.5× as often as rejected ones; DPO's samples go from 3% to 12-25% Chinese as β falls, where PPO at β = 0.2 had none and PPO at β ≤ 0.05 switched completely.
> - **Overfitting 64 pairs with DPO (β = 0.1) shows both pathologies the ladder names.** Loss 1e-6, margin 153 nats, bought on the rejected side (−167); the chosen log-ratio rises to +15, then falls through zero and ends at −14 on the training pairs and −62 on held-out chosen responses; KL to the reference 42 nats, 32% of samples in Chinese, held-out pair accuracy 0.57. The chosen side falls on the full set too: −31 after one epoch at β = 0.1.
> - **IPO stops exactly at its target margin and never collapses** (50.2 ± 0.3 for a target of 50, chosen side +20.6, KL 2.2 nats). **But DPO with a larger β stops too**: its gradient vanishes once β·h ≈ 15-25, so β = 0.5 and 1 end at margins of 32 and 25 with KL 1.6 and 0.8 and the chosen side up. The runaway is a small-β effect.
> - **On the full set IPO and DPO trace the same frontier** (IPO τ = 0.03: gold 0.952 at 8.1 nats; DPO β = 0.2: 0.957 at 9.6): in one epoch no pair reaches IPO's target, so the bound never binds. IPO's bound fixes repeated passes over few pairs, which is where DPO breaks.

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1. The task

`ladder.md` states three items:

1. Build or synthesise a preference dataset; train DPO; compare against the best PPO run.
2. Overfit DPO on a small preference set; show chosen-logprob collapse and KL blowup.
3. Swap in IPO's bounded objective; compare stability.

2. The goal

B1 maximised the KL-regularised reward

$$\max_{\pi} \; \mathbb{E}_{x,\,y\sim\pi}\big[r(x,y)\big] - \beta\,\mathrm{KL}\big(\pi(\cdot\mid x)\,\|\,\pi_{\mathrm{ref}}(\cdot\mid x)\big)$$

with a sampling loop. This objective has a closed-form maximiser,

$$\pi^*(y\mid x) = \frac{1}{Z(x)}\,\pi_{\mathrm{ref}}(y\mid x)\,e^{r(x,y)/\beta}, \qquad \text{so} \qquad r(x,y) = \beta\log\frac{\pi^*(y\mid x)}{\pi_{\mathrm{ref}}(y\mid x)} + \beta\log Z(x).$$

Any reward is therefore a policy in disguise. If preferences follow the Bradley-Terry model, $P(y_w \succ y_l) = \sigma\big(r(x,y_w) - r(x,y_l)\big)$, the partition function cancels in the difference, and fitting the reward by maximum likelihood becomes fitting the policy directly. With

$$h_\theta = \log\frac{\pi_\theta(y_w\mid x)}{\pi_{\mathrm{ref}}(y_w\mid x)} - \log\frac{\pi_\theta(y_l\mid x)}{\pi_{\mathrm{ref}}(y_l\mid x)},$$

the losses are

$$\mathcal{L}_{\mathrm{DPO}} = -\log\sigma(\beta\,h_\theta), \qquad \mathcal{L}_{\mathrm{IPO}} = \Big(h_\theta - \frac{1}{2\tau}\Big)^2.$$

No reward model, no sampling, no value function: a classification loss over pairs, with the reference's log-probabilities computed once.

The shortcut has a cost the derivation hides. Bradley-Terry needs $P(y_w \succ y_l) < 1$ to give a finite reward difference. When a pair is labelled the same way every time, the maximum-likelihood h is infinite, and DPO's loss keeps falling as h grows, whatever β is: the KL term that anchored B1's policy exists in the derivation, not in the loss being minimised. (In practice the pull is finite: the gradient of $-\log\sigma(\beta h)$ is $\beta\,\sigma(-\beta h) \approx \beta e^{-\beta h}$, which vanishes once βh is a few tens, so the margin DPO reaches scales like $1/\beta$. §7 measures where.) The loss also constrains only the *difference* of two log-ratios, so it can be lowered by pushing the rejected response down without lifting the chosen one. IPO (Azar et al., 2023) replaces the

log-sigmoid with a squared distance to a finite target margin $1/(2\tau)$, so a pair that is already separated by that much stops pulling. Items 2-3 ask whether those failure modes show up, and whether IPO's bound actually removes them.

3. What was done

Task, policy, judges: B1's, unchanged. Continue the opening 4-8 tokens of an IMDB review with up to 48 sampled tokens, positively. Qwen2.5-0.5B (base) with LoRA (r = 16, all linear layers); the reference is the same model with the adapters off. Evaluation on B1's 64 held-out test-split openings, sampled with B1's fixed seed: B1's reward model score (the Qwen head PPO trained against), the gold judge's P(positive) (`siebert/sentiment-roberta-large-english`, never trained against), KL to the reference (sum over completion tokens of $\log \pi - \log \pi_{\text{ref}}$ on the policy's own samples), distinct-2, reference NLL and the share of samples containing Chinese (PPO's hack in B1).

The preference set (`make_prefs.py`). For each of 4,096 train-split openings, four continuations sampled from the reference policy at temperature 1, each scored by B1's reward model; the pair is the best- and worst-scored of the four (`--pair-rule random` takes two at random instead). 512 test-split openings give held-out pairs the same way. These are synthetic preferences from the same reward model PPO optimised, so the comparison with PPO is about the optimiser, not the labels: PPO saw the reward through 4,800 online rollouts, DPO through four fixed reference samples per prompt. Labels are deterministic (the higher score always wins), which is the regime the IPO paper argues DPO handles worst.

Training (`dpo_lm.py`, written out rather than through TRL so every log-probability is visible). Sequence log-probabilities summed over completion tokens; batches of 16 pairs; AdamW with 10 warm-up updates; fp16 autocast on Kaggle T4s. Logged every update on the training batch, and every 25 updates on 256 held-out pairs: loss, preference accuracy ($h > 0$), h, the chosen and rejected log-ratios to the reference, and the chosen and rejected log-probabilities per token.

Runs. Full set: DPO β ∈ {0.05, 0.1, 0.2, 0.5} at lr 5e-5 (3 seeds at β = 0.1 and 0.2), β = 0.1 at lr 1e-5, with random pairs and for 3 epochs; IPO τ ∈ {0.01 (3 seeds), 0.03, 0.1}. Small set: the first 64 training pairs, 200 epochs (800 updates), DPO β = 0.1 (3 seeds), 0.5, 1; IPO τ = 0.01 (3 seeds), 0.1 (3 seeds), 1. 29 runs plus the preference set: four Kaggle notebooks, 2.2 hours of notebook time on two T4s each (≈4.5 T4-hours). B1's PPO runs supply the comparison rows.

Before training, every run evaluates the same: reward-model score −0.10, gold 0.669, distinct-2 0.88, reference NLL 3.32, 2 of 64 samples (3%) containing Chinese.

4. The preference set

split	prompts	RM score, all samples	gold, all samples	RM, best of 4	gold, best of 4	RM, worst of 4	gold, worst of 4	median RM margin best-worst	gold agrees best ≥ worst
train	4096	-0.21	0.592	1.20	0.734	-1.54	0.435	2.37	0.786
test	512	-0.13	0.603	1.28	0.735	-1.50	0.453	2.30	0.783

Wall clock 5.3 min on one T4; 0 test prompts coincide with B1's 64 evaluation prompts.

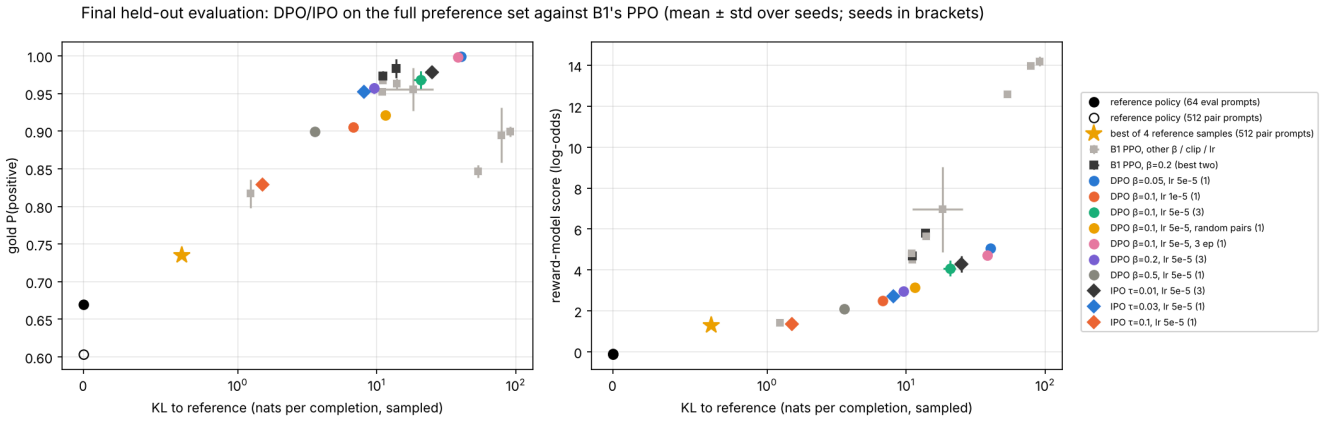
- **The labels are noisy against the gold judge**: it agrees with the reward model on which of best and worst is more positive for 78-79% of prompts. The reward model is the only teacher here, so DPO is learning *its* preferences, not the gold judge's, exactly as PPO did.
- **Best-of-4 is a weak policy**: gold 0.735 against 0.603 for a single sample (on these 512 prompts), at a KL of at most $\log 4 - 3/4 = 0.64$ nats. Everything DPO reaches beyond that is extrapolation along the direction the pairs define.
- **B1's hack is in the data, weakly.** 6.5% of chosen and 2.6% of rejected training responses contain Chinese (3.6% of all samples): the reference model drifts into Chinese on its own now and then, and the reward model scores those samples +0.86 on average against −0.25 for the rest.

5. DPO against PPO (item 1)

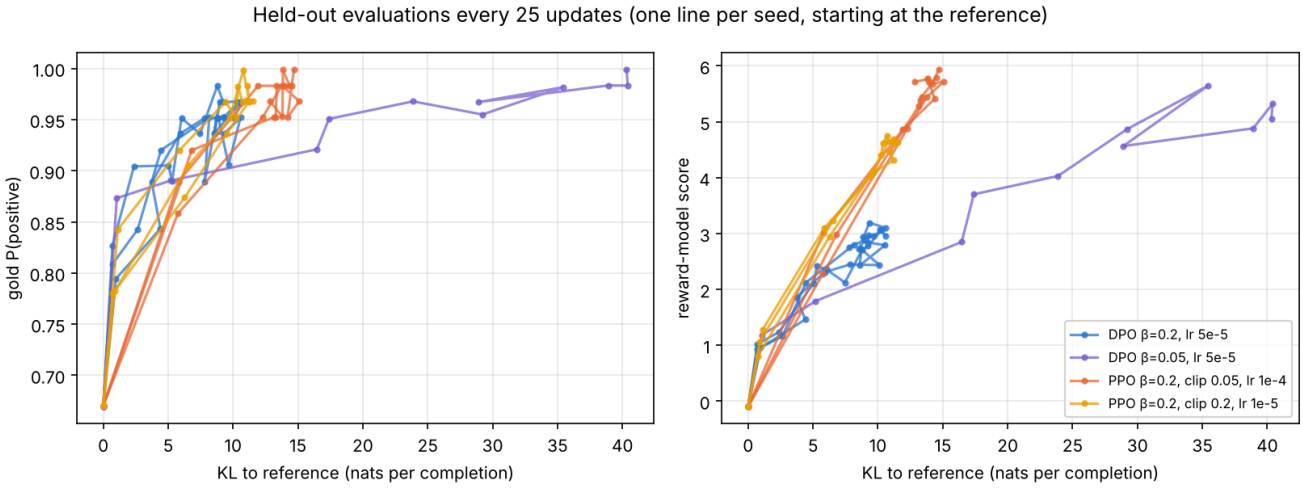
Final held-out evaluation after one epoch (256 updates of 16 pairs) unless stated, mean ± std over seeds; B1's two best PPO configurations below:

method	seeds	RM score	gold P(positive)	KL to reference	distinct-2	reference NLL	samples in Chinese	held-out pair accuracy	chosen log-ratio (held out)	rejected log-ratio (held out)	minutes per run (T4)
DPO β=0.05, lr 5e-5	1	5.05	0.999	40.32	0.87	4.11	0.25	0.83	-84.96	-129.11	5
DPO β=0.1, lr 1e-5	1	2.50	0.905	6.76	0.86	3.27	0.11	0.80	-6.47	-12.60	6
DPO β=0.1, lr 5e-5	3	4.07 ± 0.39	0.968 ± 0.012	20.70 ± 2.15	0.88 ± 0.02	3.76 ± 0.12	0.15 ± 0.03	0.83 ± 0.00	-30.84 ± 3.37	-51.45 ± 4.50	5
DPO β=0.1, lr 5e-5, random pairs	1	3.14	0.921	11.58	0.85	3.44	0.09	0.70	-15.12	-20.33	5
DPO β=0.1, lr 5e-5, 3 ep	1	4.71	0.998	38.49	0.87	3.35	0.19	0.85	-86.36	-136.79	12
DPO β=0.2, lr 5e-5	3	2.96 ± 0.02	0.957 ± 0.007	9.59 ± 0.72	0.89 ± 0.00	3.65 ± 0.00	0.12 ± 0.01	0.84 ± 0.01	-10.41 ± 0.50	-20.51 ± 0.80	5
DPO β=0.5, lr 5e-5	1	2.10	0.899	3.59	0.88	3.53	0.09	0.84	-2.12	-6.21	5
IPO τ=0.01, lr 5e-5	3	4.30 ± 0.40	0.978 ± 0.007	24.86 ± 2.64	0.86 ± 0.01	3.77 ± 0.13	0.10 ± 0.04	0.82 ± 0.01	-43.77 ± 5.09	-69.20 ± 7.58	5
IPO τ=0.03, lr 5e-5	1	2.73	0.952	8.08	0.88	3.63	0.14	0.86	-8.35	-16.96	5
IPO τ=0.1, lr 5e-5	1	1.35	0.829	1.51	0.88	3.51	0.08	0.81	-0.72	-3.10	5

method	seeds	RM score	gold P(positive)	KL to reference	distinct-2	reference NLL	samples in Chinese	held-out pair accuracy	chosen log-ratio (held out)	rejected log-ratio (held out)	minutes per run (T4)
B1 PPO β=0.2, clip 0.05, lr 1e-4	3	5.78 ± 0.11	0.983 ± 0.013	13.84 ± 0.75	0.77 ± 0.01	3.19 ± 0.01	–	–	–	–	54
B1 PPO β=0.2, clip 0.2, lr 1e-5	3	4.68 ± 0.05	0.973 ± 0.007	11.11 ± 0.33	0.80 ± 0.01	3.17 ± 0.02	–	–	–	–	56



Final evaluations on the KL axis (log scale past 1 nat). Left: the gold judge. Right: the reward model the pairs were labelled with. Grey squares are all of B1's PPO configurations, the dark ones its two best.



The same plane during training, every 25 updates, for DPO at β = 0.2 and 0.05 and the two best PPO configurations.

What the policy writes on B1's held-out prompt "I really like this movie. I":

method	completion	RM	gold
DPO β = 0.2	"…only now bought the DVD because the silent version of this is better on DVD.]]\nThe film masterful exploring of the sexuality between slaves and whites in Louisiana…"	4.7	0.999
DPO β = 0.05	"…'ve always loved the intensity and the romance of it. Ms. Butler is stunningly talented and masterful herself and sends to the highest level of excellence as well. Crystal exquisite and timeless. All great!…"	3.3	0.999

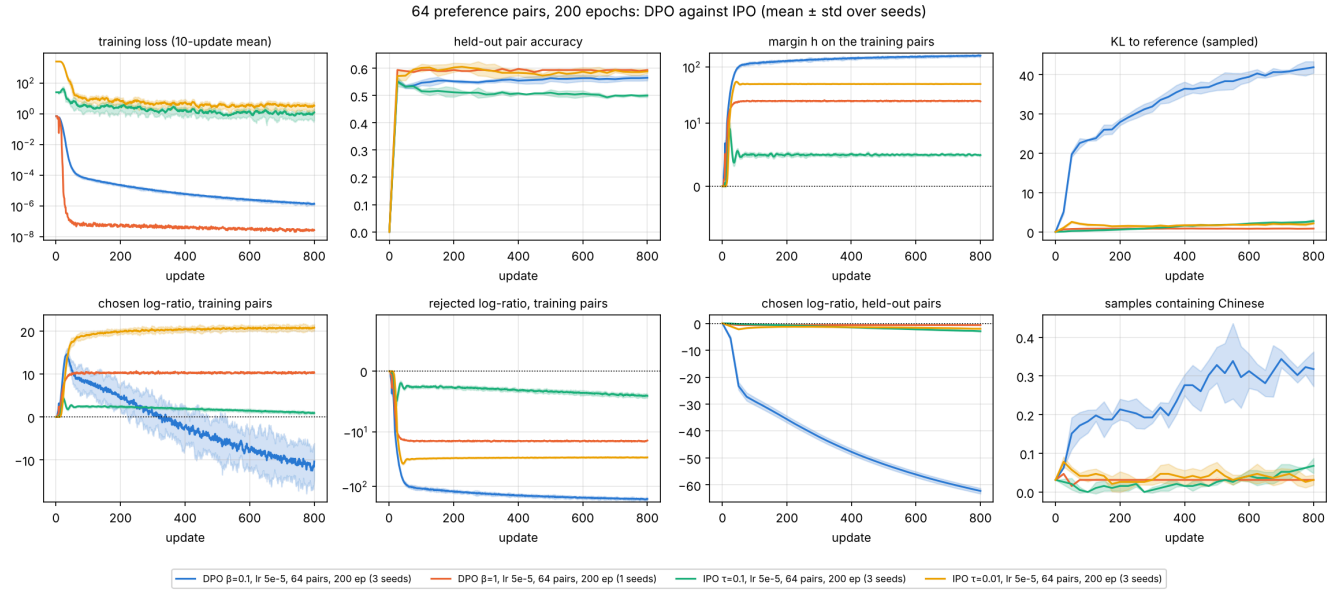
method	completion	RM	gold
IPO τ = 0.1	"…only know a little about Roman holidays, but we should learn more about them…"	4.0	0.999
PPO β = 0.2 (B1)	"…only know a little about the country, but it should appeal to the wide audience without spoiling it…"	5.4	–

- **Quality: the same gold score at the same KL, slightly lower at the operating point.** DPO β = 0.2 ends at gold 0.957 ± 0.007 and 9.6 nats; PPO β = 0.2 at 0.973 ± 0.007 and 11.1 nats (lr 1e-5) or 0.983 ± 0.013 and 13.8 nats (lr 1e-4). Along the trajectories (left panel) DPO's and PPO's gold-against-KL curves overlap; the gap at the end is where each stops, not a worse curve. DPO also keeps more diversity: distinct-2 0.89 against 0.77-0.80 for PPO.
- **PPO buys reward-model score that the gold judge does not credit.** At ≈10 nats PPO's reward-model score is 4.7 and DPO's 3.0 (right panels), for the same gold. PPO samples online and climbs the reward model wherever it is steepest; DPO only sees four reference samples per prompt, so it can only move along directions those samples span. That is less optimisation against the proxy, which here costs nothing on the judge that matters.
- **β sets the KL budget, as the derivation says.** β = 0.5, 0.2, 0.1, 0.05 end at 3.6, 9.6, 20.7, 40.3 nats, and gold rises with it (0.90 → 0.999). The learning rate moves along the same curve (β = 0.1 at lr 1e-5: 6.8 nats, gold 0.905), and so does a third epoch (β = 0.1: 38.5 nats, gold 0.998).
- **The reward model's exploit arrives offline, at the strength it has in the data.** 12% of DPO's samples at β = 0.2 contain Chinese (3% before training), 15% at β = 0.1 and 25% at β = 0.05. PPO at β = 0.2 had none, and at β ≤ 0.05 switched completely within 25 updates. DPO cannot discover the exploit, but it does learn the preference for it that the labels carry, and a lower β lets it act on that more.
- **The chosen response gets *less* likely even here.** Held-out chosen log-ratios are negative for every full-set configuration (−10 at β = 0.2, −31 at β = 0.1, −85 at β = 0.05), with the rejected ones falling faster; the preference is learned (held-out pair accuracy 0.82-0.86) as the difference of two falls. Section 6 shows what that becomes when nothing stops it.
- **Engineering cost.** A DPO run is ≈5 minutes on one T4 including 11 evaluations, against ≈55 for PPO; it holds one model with adapters in training (the reward model and gold judge are loaded for evaluation only), where PPO holds the policy, a value model, the reward model and the reference. The preference set costs 5 minutes once. PPO's settings that had to be found in B1 (β, clip range, learning rate, which value-model layers fit in memory) become two for DPO (β and the learning rate), and the pairing rule: random pairs (median reward gap 1.10 against 2.37) end at 11.6 nats and gold 0.921 for the same β = 0.1.

6. Overfitting a small set: DPO (item 2)

The first 64 training pairs, 200 epochs (800 updates of 16 pairs), lr 5e-5:

method	seeds	train loss	train accuracy	train margin h	chosen log-ratio (train)	lowest chosen log-ratio (train)	rejected log-ratio (train)	held-out accuracy	KL to reference	peak KL	gold P(positive)	distinct-2	samples in Chinese
DPO β=0.1, lr 5e-5, 64 pairs, 200 ep	3	1.22e-06 ± 1.02e-07	1.00 ± 0.00	153.3 ± 5.2	-13.9 ± 3.9	-31.17 ± 1.92	-167.2 ± 4.1	0.57 ± 0.01	41.9 ± 1.5	41.98 ± 1.52	0.837 ± 0.029	0.85 ± 0.01	0.32 ± 0.04
DPO β=0.5, lr 5e-5, 64 pairs, 200 ep	1	3.47e-07	1.00	32.5	11.5	0.00	-21.0	0.57	1.6	1.56	0.661	0.89	0.05
DPO β=1, lr 5e-5, 64 pairs, 200 ep	1	2.97e-09	1.00	24.5	9.1	0.00	-15.4	0.59	0.8	0.89	0.760	0.89	0.03
IPO τ=0.01, lr 5e-5, 64 pairs, 200 ep	3	1.72 ± 1.17	1.00 ± 0.00	50.2 ± 0.3	20.6 ± 0.6	-0.02 ± 0.02	-29.6 ± 0.9	0.59 ± 0.01	2.2 ± 0.1	2.53 ± 0.21	0.672 ± 0.013	0.90 ± 0.00	0.03 ± 0.01
IPO τ=0.1, lr 5e-5, 64 pairs, 200 ep	3	1.55 ± 0.914	1.00 ± 0.00	5.1 ± 0.2	1.5 ± 0.3	-0.09 ± 0.09	-3.7 ± 0.3	0.50 ± 0.01	2.7 ± 0.4	2.75 ± 0.37	0.631 ± 0.009	0.91 ± 0.00	0.07 ± 0.02
IPO τ=1, lr 5e-5, 64 pairs, 200 ep	1	1.12	0.69	0.2	-0.7	-1.89	-0.9	0.50	2.1	2.13	0.579	0.91	0.03



Mean ± std over seeds (one seed for DPO β = 1). Training-pair panels are 10-update running means.

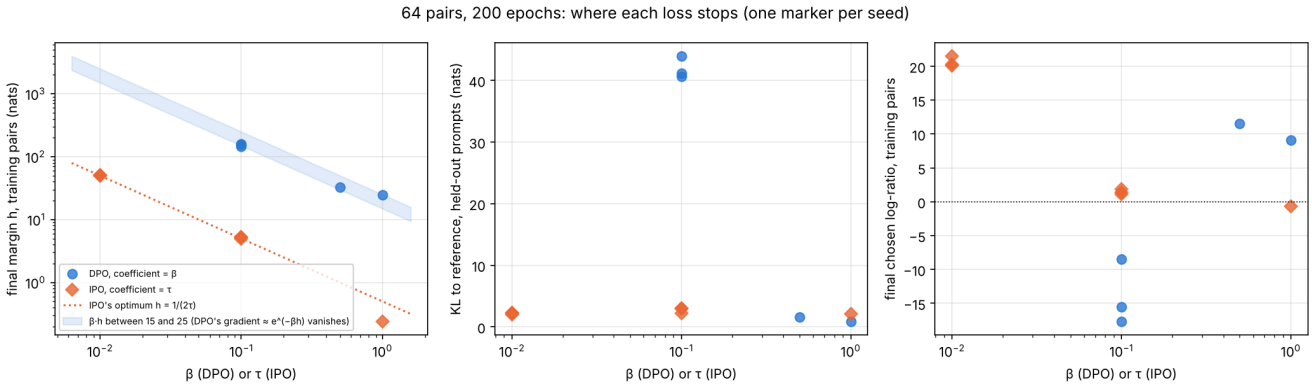
What DPO β = 0.1 writes after 200 epochs on 64 pairs (seed 0):

> "Of all the films I've看过, I really like Inception. It's迷人又迷人的文学 translated into motionpicture…"

- **The aggregate loss says everything is fine.** It falls to 1e-6 and every training pair is ranked correctly from update 5 on.

- **Underneath, the chosen side collapses.** The margin reaches 153 nats, bought almost entirely by pushing the rejected responses down (−167). The chosen log-ratio first rises (+14 to +16 at update 26-31), then turns, crosses zero at update 255-315 on all three seeds and ends at −14 (−0.2 nats per token): after training the model is *less* likely than before to write the responses it was told to prefer. On held-out chosen responses the fall is monotone, to −62.
- **KL blows up and the policy drifts where the labels lean.** Sampled KL to the reference rises to 42 nats; 32% of samples contain Chinese (the lean of §4, amplified); gold rises to 0.84 and held-out pair accuracy is 0.57: the 64 pairs were memorised, not generalised from.
- **Why the chosen side falls (a hypothesis, not tested here).** The loss only sees h, the difference of two log-ratios; nothing in it asks for the chosen side to stay up. Chosen and rejected continuations of the same opening are written in the same register by the same model, so the cheapest direction that lowers the rejected sequence's probability plausibly lowers the chosen one's too, just less; once the pairs are separated every step still increases h, so the shared fall accumulates.

7. IPO's bound (item 3)



The 64-pair runs, final values, one marker per seed: the margin each loss stops at (left), KL to the reference (middle), and the chosen log-ratio on the training pairs (right), against DPO's β or IPO's τ.

- **IPO stops exactly where τ says**: final margins 50.2 ± 0.3 for τ = 0.01 (target 1/(2τ) = 50) and 5.1 ± 0.2 for τ = 0.1 (target 5); τ = 1 (target 0.5) is still short at 0.2 after 800 updates.
- **IPO reaches a large margin without the collapse.** At τ = 0.01 it separates the 64 pairs by 50 nats with the chosen side *up* 20.6 and the rejected down 29.6, at 2.2 nats of KL; DPO β = 0.1 reached 153 nats at 42 nats of KL with the chosen side down 14. No IPO run collapses on any seed, and its Chinese share stays at the reference's 3-7%.
- **But DPO with a larger β stops too, and just as cleanly.** Its gradient, $\beta\sigma(-\beta h) \approx \beta e^{-\beta h}$, vanishes once $\beta h \approx$ 15-25 (the shaded band): DPO stops at margins of 153, 32.5 and 24.5 for β = 0.1, 0.5 and 1. At β = 0.5 and 1 (one seed each) the chosen side ends up (+11.5, +9.1) and KL is 1.6 and 0.8 nats. The runaway of §6 is the bound sitting at

≈20/β nats, far enough at β = 0.1 to wreck the policy. What IPO adds is a stopping point set explicitly by τ, instead of one set by where an exponential tail becomes numerically negligible.
- **On the full set the bound never binds.** IPO τ = 0.03 (target 16.7) ends at 8.1 nats and gold 0.952; τ = 0.01 (target 50) at 24.9 nats and gold 0.978 ± 0.007, with a held-out chosen log-ratio of −44, the same fall DPO shows. One pass over 4,096 pairs leaves every pair short of its target, so IPO behaves like DPO with a re-labelled coefficient and lands on the same frontier (`frontier.png`).
- **Stability, compared:** where DPO fails, on repeated passes over few pairs at small β, IPO does not; where DPO does not fail, IPO changes nothing. With one epoch, or with β ≥ 0.5, DPO did not need the fix.

8. What this sets up

- **B3 (GRPO)** returns to online sampling. The comparison here says what online sampling buys on this task: more reward-model score at the same KL, none of it credited by the gold judge. A verifiable reward (B3's RLVR) removes the proxy, which is the case where online optimisation should pay.
- **B4 (overoptimisation)** gets a second route to the same exploit: offline, through labels that lean towards it, at the strength of the lean. A pair filter on language, or a reward-model ensemble used for labelling, is a mitigation that applies to DPO's data rather than to its loss.
- **B5 (distillation)** can take DPO β = 0.2 as a cheap teacher candidate: PPO's gold score at a tenth of the cost, with more diverse samples.
- **Reading the curves:** DPO's loss and pair accuracy are not the result. The chosen log-ratio, the sampled KL and the samples are; the loss looked perfect in every run where the policy collapsed.

9. Deviations and what went wrong

- **DPO and IPO are written out** (`dpo_lm.py`) rather than run through TRL's DPOTrainer, so every log-probability that items 2-3 ask about is logged at every update. IPO uses summed sequence log-probabilities, as in the paper's derivation; TRL's IPO averages per token, which rescales τ.
- **Preferences are synthetic and deterministic**, from B1's reward model, not human. That makes the PPO comparison about the optimiser, and it is the regime (labels that never flip) where the IPO paper expects DPO to overfit; with noisy or human labels the picture can differ.
- **Single seeds** for DPO β = 0.05, 0.5 and 1, IPO τ = 0.03, 0.1 (full set) and τ = 1, and the random-pair and 3-epoch runs; the claims that rest on them (DPO's margin at large β, the frontier positions) are marked by the seed counts in the tables.
- **The learning rate was not tuned per loss.** All IPO and most DPO runs use lr 5e-5 (LoRA); one DPO run at 1e-5 lands on the same curve, at lower KL.
- **Kaggle's T4 queue** held two of the four notebooks for about an hour each before they started; the runs themselves took 2.2 notebook-hours.